

■ ENHANCING YOUR VIEW

The naked eye alone can be used to identify the constellations, view planets and meteors, and locate dark and light features on the Moon. However, equipment such as binoculars will enhance your view by collecting more light than the human eye can. They add clarity to your view of the Moon's surface and of stellar objects such as star clusters and the Orion Nebula, as well as revealing objects not visible to the naked eye.

FOCUSING BINOCULARS



1 Try the binoculars

Everyone's eyesight is different, so you must focus binoculars before use.



2 Locate eyepiece focus

Find which eyepiece can be focused independently. Look through with that eye closed.



3 Focus left eyepiece

Rotate the binoculars' main, central focusing ring until the left image is in sharp focus.



4 Switch eyes, and focus

Open only the other eye. Use the eyepiece focusing ring to bring the image into focus.



5 Use the binoculars

Both eyepieces should now be in focus. Open both eyes and start observing.

BINOCULARS

The most useful optical aid for a newcomer to astronomy is a pair of binoculars. They are easy to use and portable, and they show the image the right way up (unlike telescopes). They are a combination of two low-powered telescopes, using both eyes for viewing.

Binoculars come in a range of sizes. The two numbers that describe a pair of binoculars are important. The first is magnification; the second, the diameter of the objective lens. Binoculars

described as 7 x 50 magnify an object seven times and collect the starlight with a lens 50mm wide. Larger binoculars, which have a lens of 70mm or more and magnifications of 15–20, are used by dedicated astronomers. Binoculars are difficult to keep steady, whatever the size. For a stable image with handheld binoculars, sit down, or lean against a low wall, resting the binoculars on top.



hold steady sitting and supporting arms on knees

MOON THROUGH STANDARD BINOCULARS



ANDROMEDA GALAXY (M31)



Large binoculars

Too heavy to hold still, large binoculars should be supported on a tripod. The larger lens and higher magnification reveal more than standard binoculars.



MOON THROUGH LARGE BINOCULARS



ANDROMEDA GALAXY (M31) THROUGH LARGE BINOCULARS

GETTING STARTED

■ GETTING EVEN CLOSER

Telescopes bring astronomical objects even closer than binoculars. They gather more light and make faint objects seem brighter and larger.

TELESCOPES

There are two types of telescopes: refractors, which uses lenses; and reflectors, which use mirrors. Both produce upside-down images, but this is only noticeable when looking at familiar objects. A telescope is further described by the diameter of its main lens or mirror— 3 inch (75mm), for example. The diameter, known as the aperture, is important. Double this, and the light gathered by the telescope quadruples. This light forms an image that is magnified by the eyepiece.

Amateur telescopes are now often controlled by a computer, simplifying the work of locating and tracking particularly faint objects.



SATURN THROUGH 3-IN (75MM) TELESCOPE



SATURN THROUGH 12-IN (300MM) TELESCOPE

Reflecting telescope

Telescopes for amateur astronomers, with an aperture over 4 in (100mm), are usually reflectors. Examples such as this 8-in (200mm) computer-controlled one (left) are easily transported to a dark site.